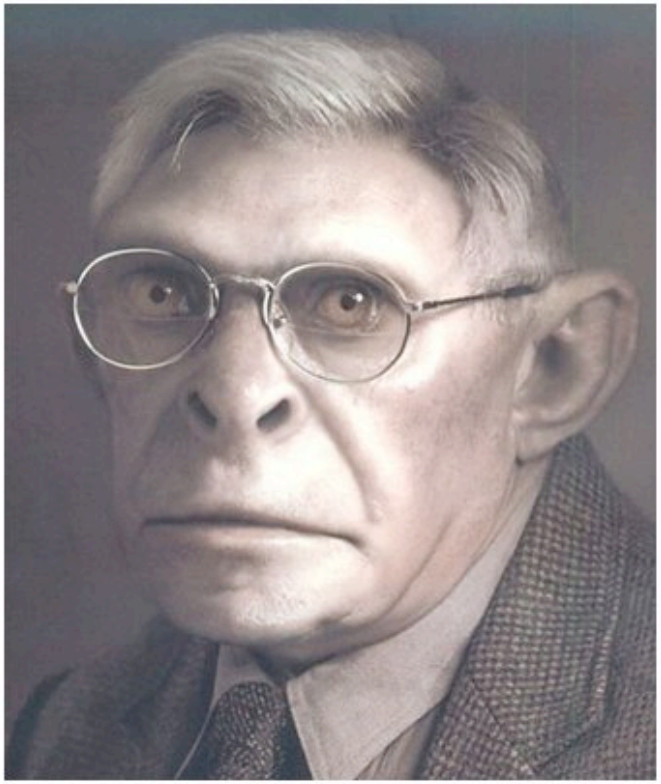
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Comparative Anatomy of Man & Apes

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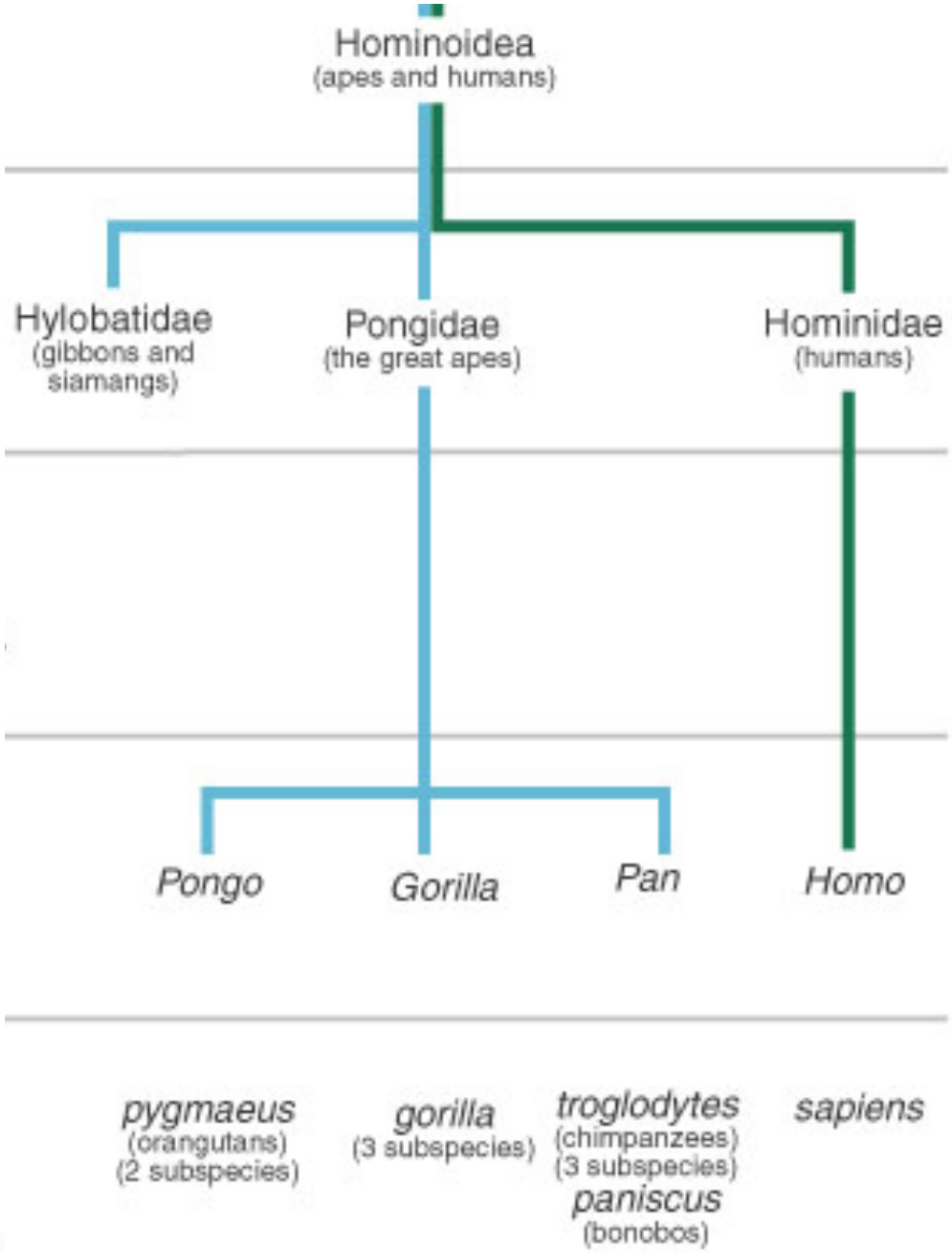
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Classification

- Apes and Men belong to Sub-order Anthropoidea
- Super Family - Hominoidea
- Gibbons - Hylobatidae
- Chimps, Orangs, Gorillas - Pongidae
- Man - Hominidae

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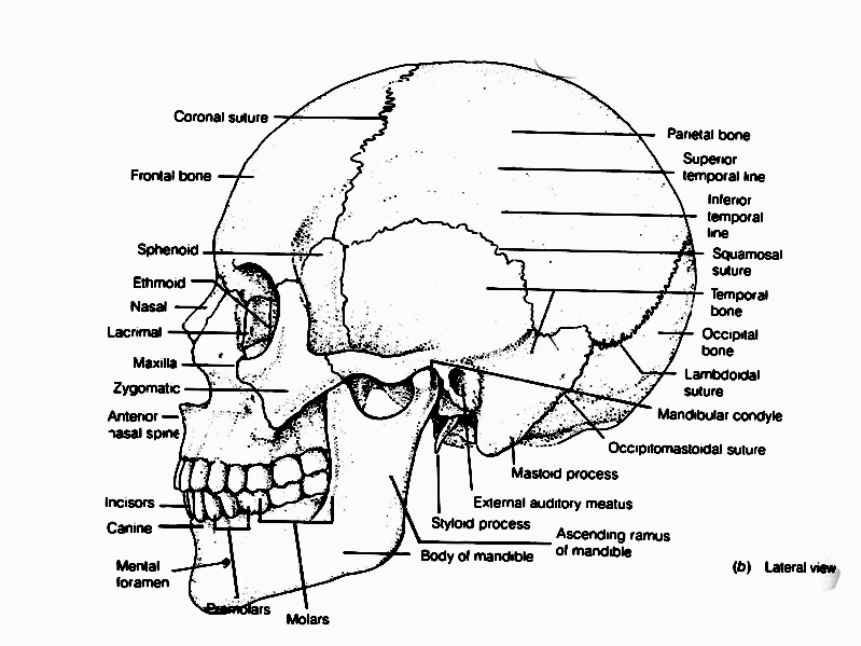
Comparative Anatomy

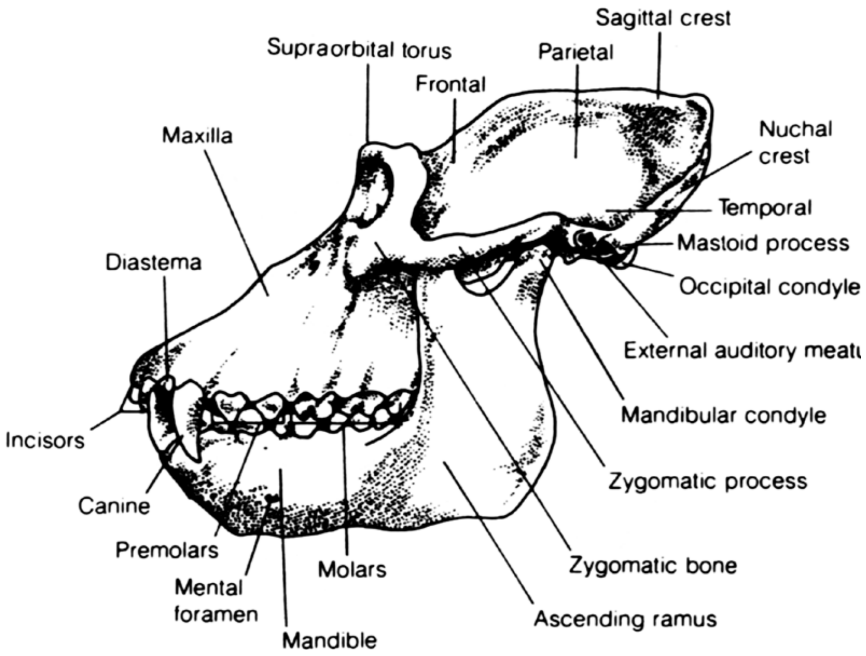
Trait	Apes	Man
Forehead	Absent	Present
Supra Orbitals	Heavy	Poor
Cranial Vault	Flat	High & Rounded
Occiput	High	Low
Foramen Magnum	Posterior	Center
Nasal Bridge	Depressed	High
Nasal Aperture	Large	Moderate to Small
Facial Profile	Dished Out	Relative Even

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Human & Ape Skulls

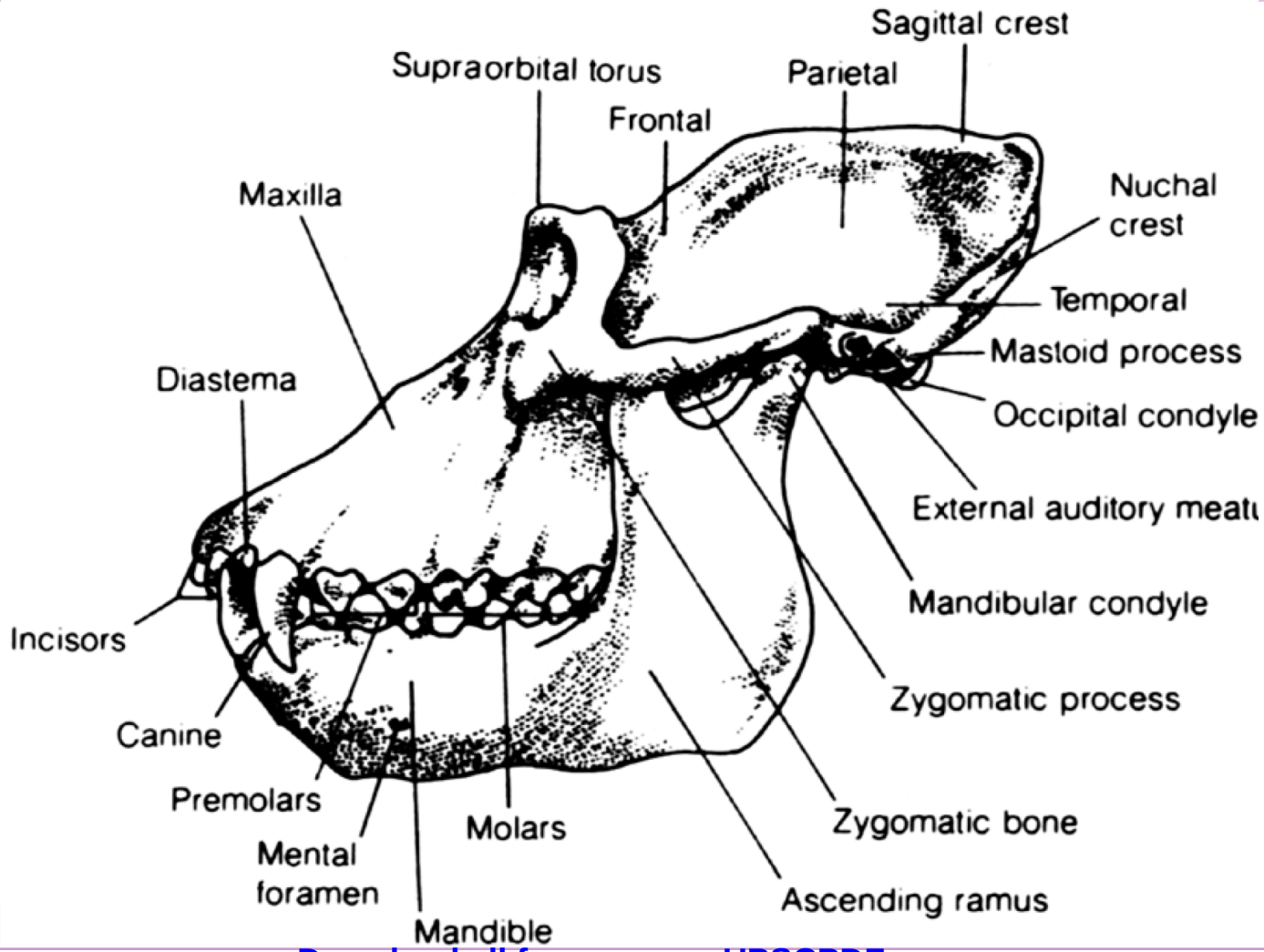




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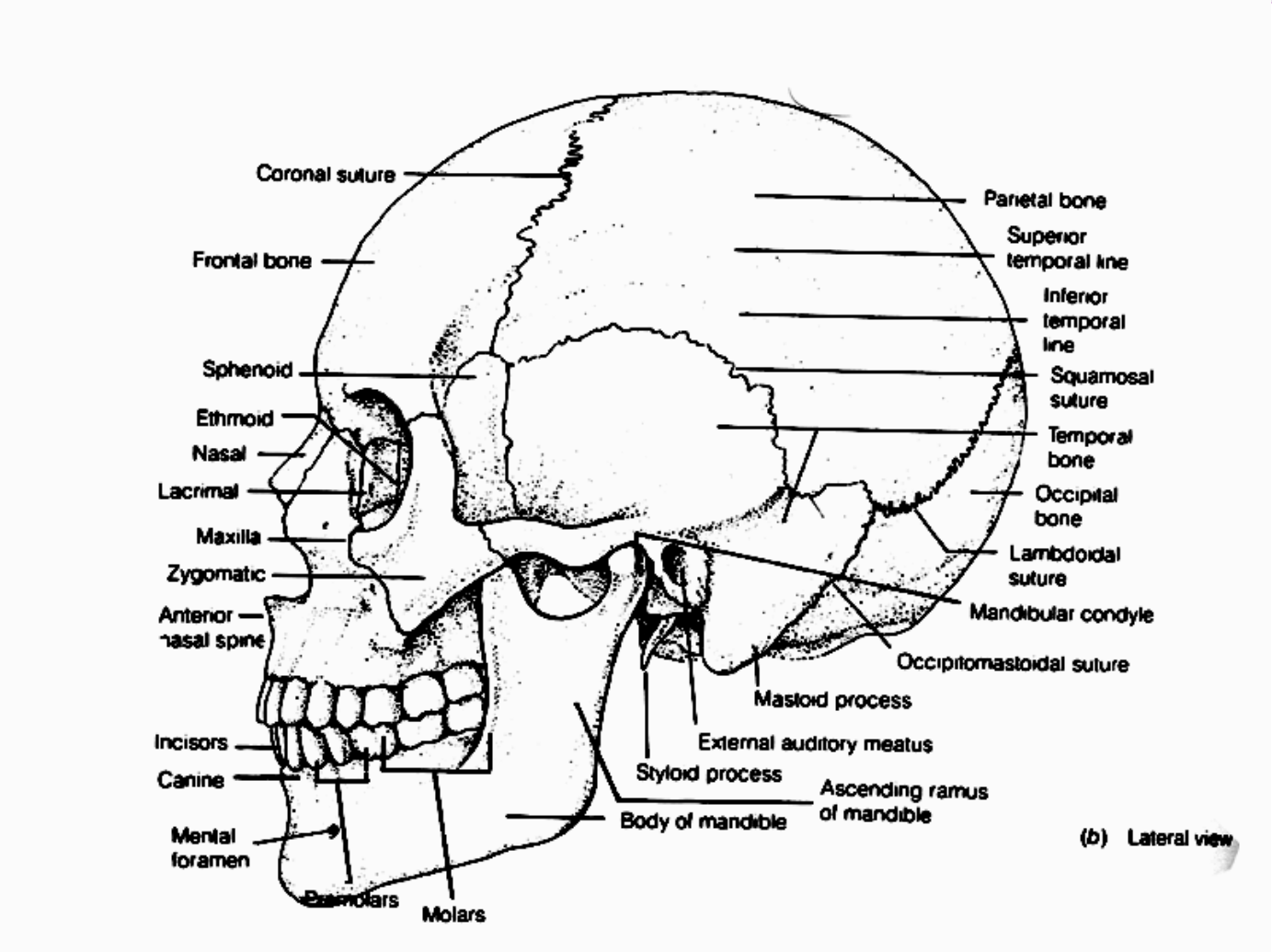
Ape's Skull



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Human Skull



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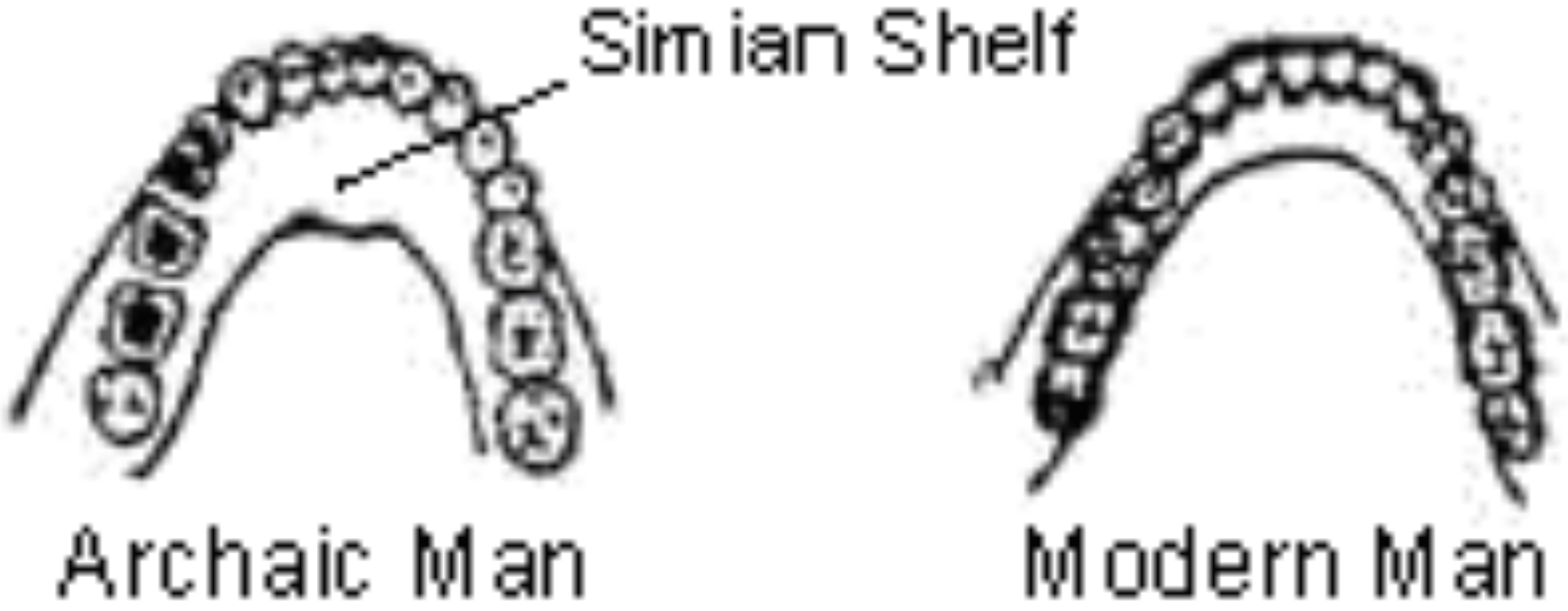
Comparative Anatomy

Trait	Apes	Man
Incisors	Large & procumbent	Small & vertical
Canines	Large & procumbent	Small & non-projecting
Diastema	Present	Absent
Dental Arcade	U-shaped	Parabolic
Jaws	Massive	Moderately small
Simian shelf (a thickening in the front of mandible)	Present	Absent
Chin	Absent	Present

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Simian Shelf



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Sagital Crest



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Comparative Anatomy

Trait	Apes	Man
Vertebral Column	Parallel to ground	Perpendicular
Head	Horizontal to bent	Upright
Lips	Stretched over bulging jaws	Thin and non-protrusive
Upper Lip	Median furrow absent	Present
Arms	Elongated with brachiating anatomy	Short
Thigh	Short, thick and curved	Long, slender & elongated
Linea aspera	Absent	Prominent
Femur cross section	Round and oval	Prismatic
Great toe	Pseudo opposable	Truly opposable

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Stature and Weight

Species	Height	Weight
Gibbon	3 Feet	6.5 Kgs
Orangutan	4 Feet 6 Inches	74 Kgs
Chimpanzee	5 Feet	45 Kgs
Gorilla	5 Feet 5 Inches	160 Kgs
Man	5 Feet 6 Inches	65.5 Kgs

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• Brain Size

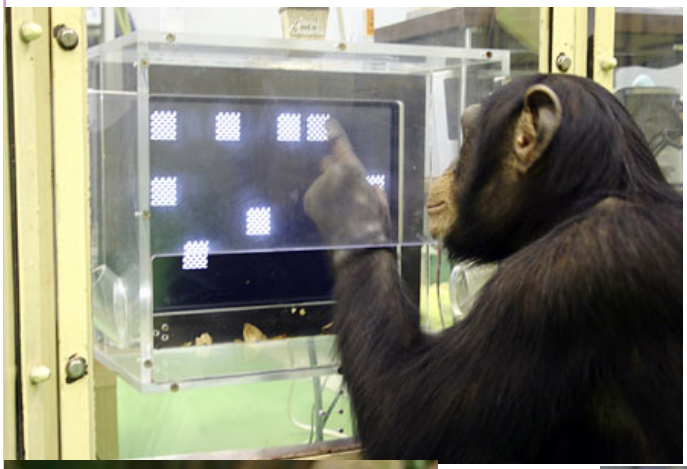
Species	Cranial Capacity
Gibbon	549 CC
Chimpanzee	400 CC
Orangutan	416 CC
Gorilla	598 CC
Humans	1450 CC

- Weight of human brain is twice as heavy as gorilla's
- Frontal region of human brain is highly developed

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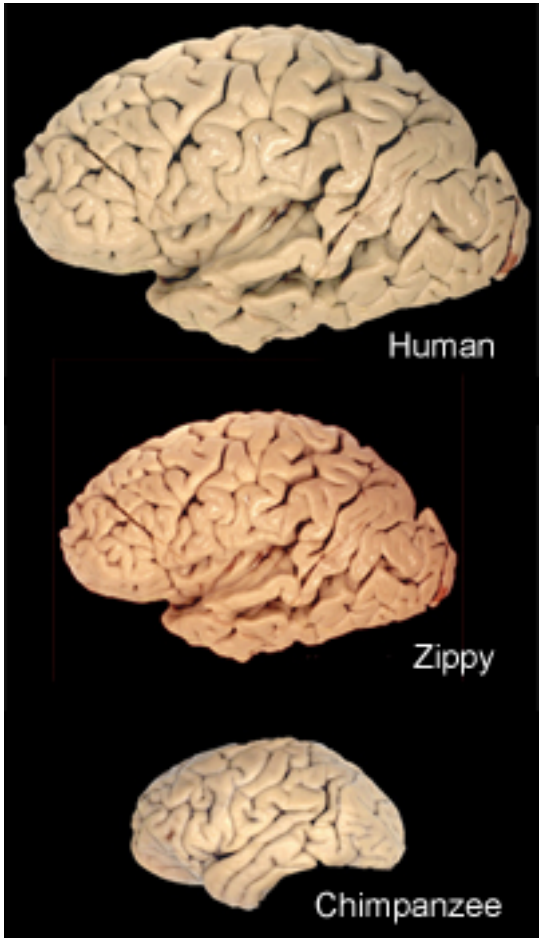
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Zippy









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Similarities?

- The differences are only of degree and not kind
- Physiologically, all the apes are same
- The structure of uterus, placenta and even the sexual rhythm in female chimps is same as human female
- Serological similarities are seen in all apes and men
- Diseases like typhoid, cholera and syphilis can be seen in apes too!

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Man's Closest Relatives?

- Our closest living relatives are chimpanzees
- 3.9% difference in DNA between *Pan troglodytes* and *Homo sapiens* (Ze Cheng *et* al. 2005.)
  - 1.2% of difference is due to single-nucleotide differences
  - 2.7% is due to duplications (mostly in chimps; responsible for many changes to chromosome structure)
  - This is 10 times smaller difference than between many other species (ex. Mice and rats)
- Other estimates range upwards to 98.4% (Diamond 1993)

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